San Francisco 49ers Offense: Scouting Report

Public data. Built from nflverse (nflfastR play-by-play, NGS participation, FTN charting), mapped onto the PFF GamePBP layout. Run concepts, PFF success codes, yards after contact and receiver alignment are not in this source; see *What nflverse cannot cover* at the end.

Sample: 8 games (Sep 10, 2026 to Nov 08, 2026) · 453 offensive plays · Opponents: ARZ, ATL, DEN, LA, LV, MIA, SEA, WAS
Source: SF_2026_nflverse_GamePBP.csv · Generated Nov 10, 2026 18:07 · Success = standard yardage definition · EPA = nflfastR model

57%	51%	+0.19	59%	53%
PASS RATE	NEUTRAL EARLY-DOWN PASS RATE	EPA / PLAY	SUCCESS RATE	3RD-DOWN CONVERSION

3%	11%
SACK RATE	EXPLOSIVE PLAY RATE

Key findings

TENDENCIES

- Passes on 57% of all plays and 51% of neutral early-down plays (1st/2nd down, outside 2-minute drills).
- QB alignment is a strong run/pass key: under center they run 65% of the time (n=233); in shotgun they pass 81% (n=214).
- Tell: Shotgun + 3rd Down → pass 90% of the time (n=61, 95% lower bound 80%).
- Tell: 2nd Down + Extra Long (11+) → pass 95% of the time (n=20, 95% lower bound 76%).
- Tell: Shotgun → pass 81% of the time (n=214, 95% lower bound 76%).
- Tell: 3rd Down + Medium (4-6) → pass 91% of the time (n=22, 95% lower bound 72%).
- Run game is built on Middle and Right Guard: 56% of carries.
- Predictability: knowing personnel, alignment, down and distance lifts a defense's run/pass guess from 57% to 72% (leave-one-out).

STRENGTHS

- Most efficient run gap: Left Tackle, +0.11 EPA/carry, 4.5 YPC, 59% success (n=17).
- Primary target: G. Kittle (#85) (TE), 23% target share, 37 catches for 457 yards, +0.25 EPA/target, 21 first downs.
- Most efficient receiver: M. Evans (#5) (WR), +1.02 EPA/target on 39 targets, aDOT 14.6, 2.7 YAC/rec.
- Lead back C. McCaffrey (#23): 52% of carries, 4.5 YPC, 57% success. Favorite gap: Middle.
- Most total value: M. Evans (#5), 39.9 total EPA on 39 touches + targets (500 scrimmage yards).
- Deep shots (20+ air yards): 9% of targets, 55% complete, +1.15 EPA/target (n=20).

VULNERABILITIES

- Play action: 23% of dropbacks, +0.08 EPA/play vs +0.37 without (n=58 vs 198).
- Least efficient run gap: Right Guard, -0.08 EPA/carry, stuffed (0 or fewer yards) on 7% (n=27).
- On 3rd and Long (7-10) they convert only 31% (n=13); they pass 92% of the time there. Overall 3rd-down conversion: 53%.
- High-volume, low-return target: C. McCaffrey (#23) (HB), 20% of targets but +0.08 EPA/target and 80% catch rate.

n = the number of plays (or carries / targets) behind a number, i.e. the sample size. Findings are generated by fixed rules (see Methodology) and include sample sizes. Treat any rate built on fewer than 10 plays as anecdotal.

1. Personnel run/pass tendencies

No plays to chart

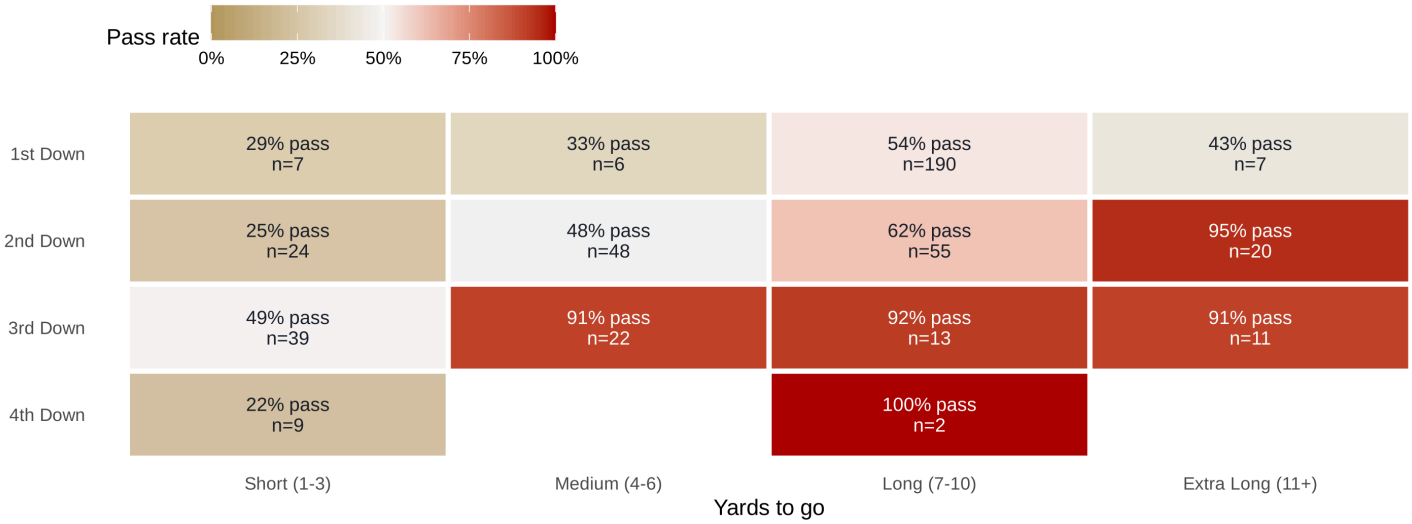
Not available for this data set.

453 plays with no recorded personnel are left out of this section.

2. Down-and-distance tendencies

Pass rate by down and distance

Each box: share of plays that were passes, and n = number of plays in that down-and-distance situation



Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Down	To go	Plays	Pass %	Run EPA	Pass EPA	Success
1st Down	Short (1-3)	7	29%	+0.24	+0.65	57%
1st Down	Medium (4-6)	6	33%	+0.11	-0.03	67%
1st Down	Long (7-10)	190	54%	-0.05	+0.16	62%
1st Down	Extra Long (11+)	7	43%	+0.26	+0.29	43%
2nd Down	Short (1-3)	24	25%	+0.09	+0.19	71%
2nd Down	Medium (4-6)	48	48%	-0.11	+0.23	56%
2nd Down	Long (7-10)	55	62%	+0.24	+0.67	58%
2nd Down	Extra Long (11+)	20	95%	-0.56	+0.43	45%
3rd Down	Short (1-3)	39	49%	+0.27	+0.18	64%
3rd Down	Medium (4-6)	22	91%	-0.64	+0.36	50%
3rd Down	Long (7-10)	13	92%	+0.00	-0.68	31%
3rd Down	Extra Long (11+)	11	91%	+0.33	+1.37	45%
4th Down	Short (1-3)	9	22%	+0.19	+2.91	78%
4th Down	Long (7-10)	2	100%	n/a	-0.23	50%

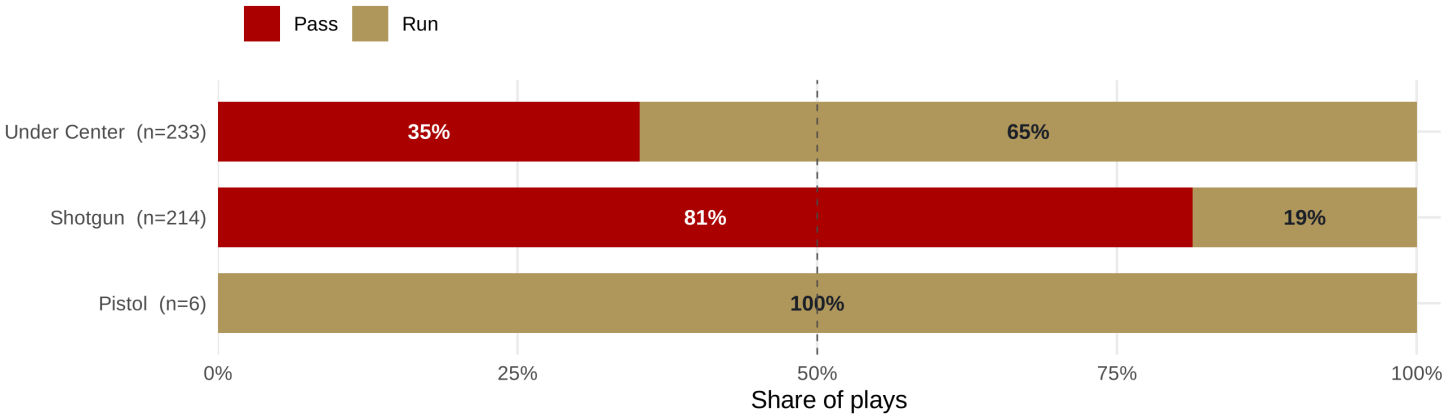
By field position (yards from the opponent's end zone):

Zone	Plays	Pass %	EPA/play	Success
Goal to Go (1-10)	44	48%	+0.41	61%
Red Zone (11-20)	42	60%	+0.53	60%
Plus Territory (21-50)	156	54%	+0.08	59%
Own Territory (51-80)	174	60%	+0.13	58%
Backed Up (81-99)	37	54%	+0.27	57%

3. QB alignment tendencies

Run/pass split by QB alignment

n = number of plays from that alignment; read small groups with caution



Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Alignment	Plays	Snap %	Pass %	Run EPA	Pass EPA	Run YPC	Pass Y/P
Under Center	233	51%	35%	+0.09	+0.27	4.3	7.5
Shotgun	214	47%	81%	-0.14	+0.32	5.2	8.3
Pistol	6	1%	0%	-0.02	n/a	3.5	n/a

Alignment by personnel (groupings with enough plays):

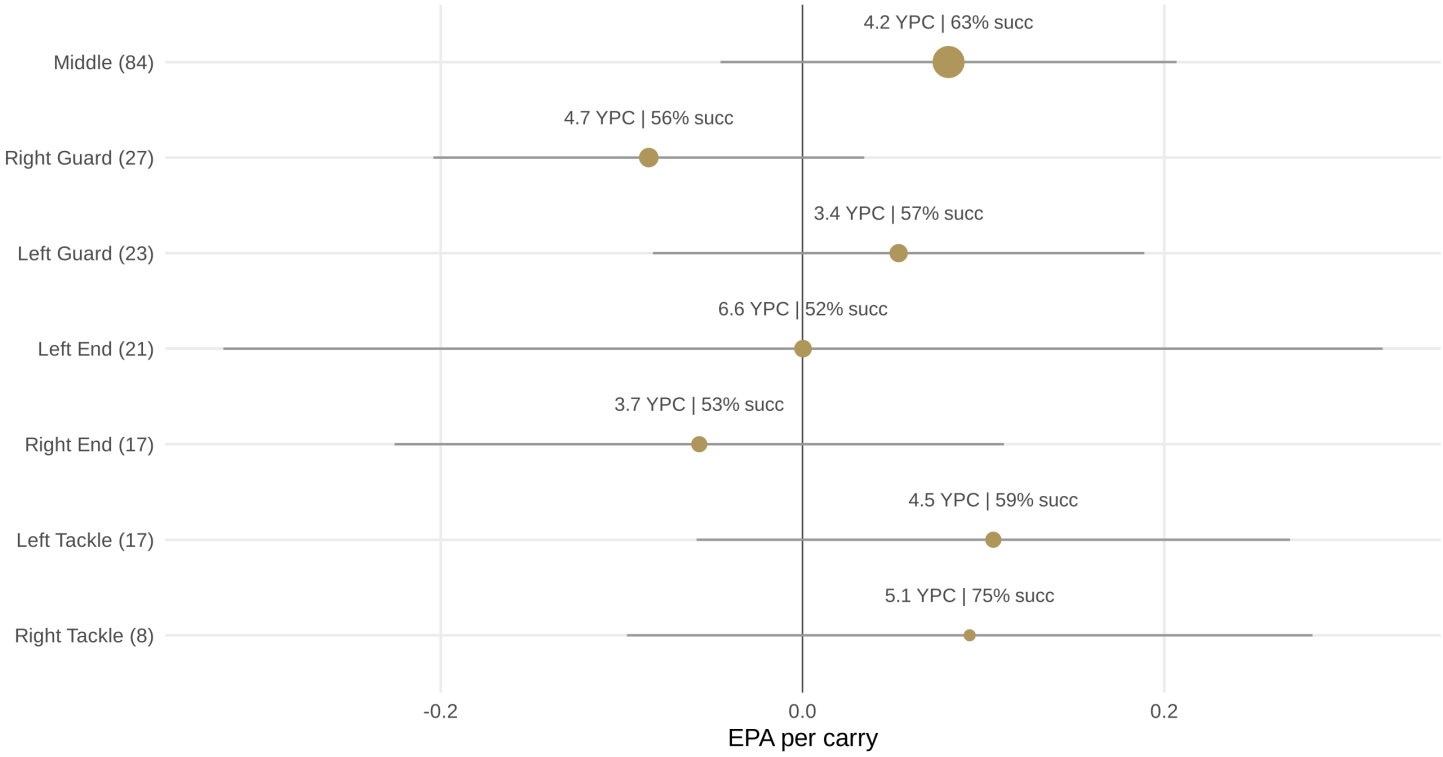
Not available for this data set.

4. Run gap and direction: frequency and effectiveness

nflverse does not chart blocking schemes, so runs are grouped by the gap and side the play-by-play records (end / tackle / guard / middle). This answers *where* they run, not *which concept*.

Run gap and direction: volume and efficiency

Ordered by carries (in brackets). Point = EPA per carry, bar = +/- 1 standard error



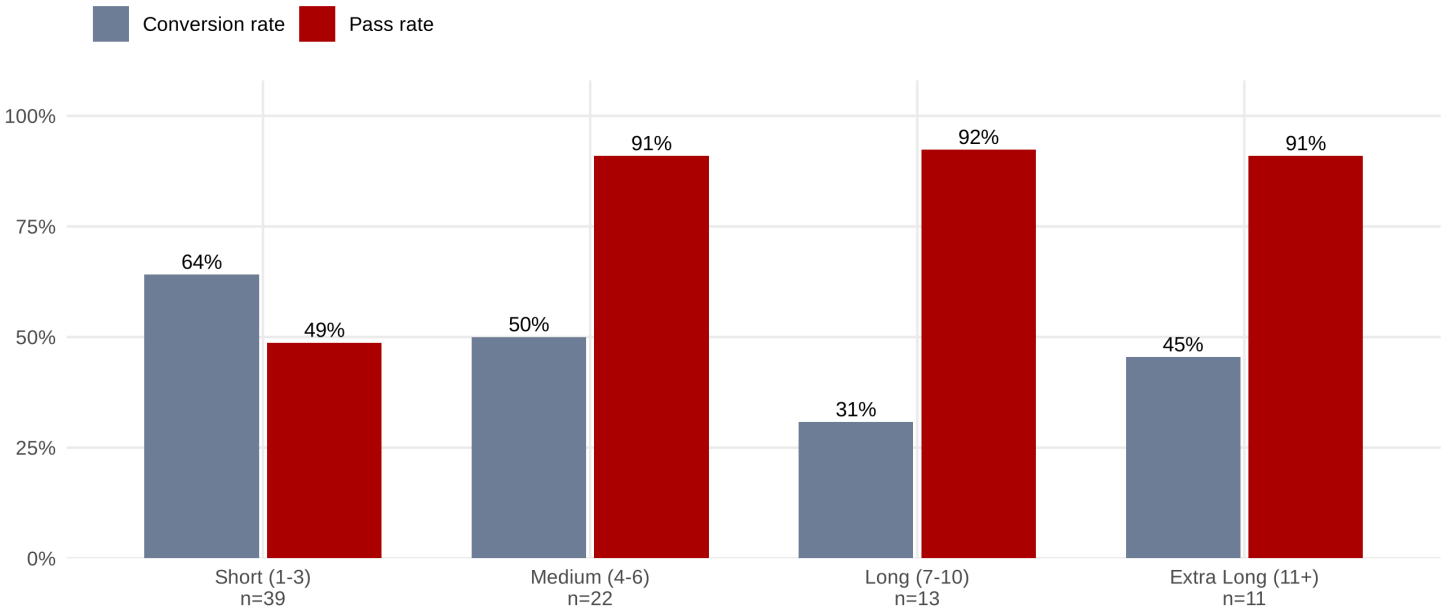
Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Concept	Carries	Share	YPC	Median	Success	EPA/carry	Stuff %	Explosive %	Under center %	Top personnel
Middle	84	43%	4.2	3.0	63%	+0.08	17%	12%	73%	Unknown
Right Guard	27	14%	4.7	3.0	56%	-0.08	7%	15%	81%	Unknown
Left Guard	23	12%	3.4	3.0	57%	+0.05	13%	4%	78%	Unknown
Left End	21	11%	6.6	6.0	52%	+0.00	24%	29%	76%	Unknown
Left Tackle	17	9%	4.5	5.0	59%	+0.11	6%	6%	76%	Unknown
Right End	17	9%	3.7	3.0	53%	-0.06	18%	6%	88%	Unknown
Right Tackle	8	4%	5.1	5.0	75%	+0.09	0%	12%	75%	Unknown

5. Third-down behavior and key targets

Third down: how often they throw and how often they convert

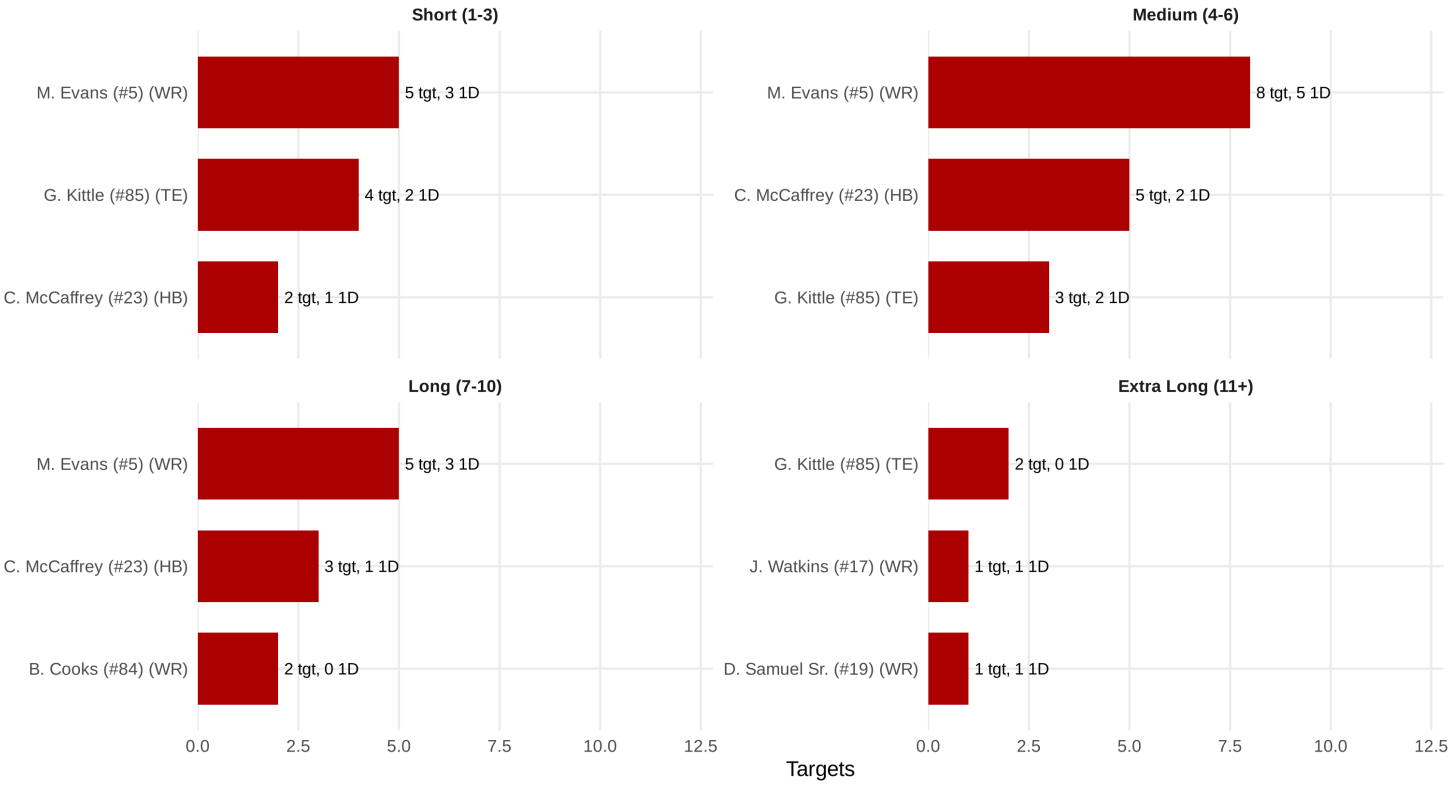
n = number of third-down plays at that distance



Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

To go	Plays	Pass %	Conv %	Conv 95% CI	EPA/play	Gun/Pistol %	Avg air yds	Sack %
Short (1-3)	39	49%	64%	48%-77%	+0.22	41%	9.6	5%
Medium (4-6)	22	91%	50%	31%-69%	+0.27	95%	9.1	0%
Long (7-10)	13	92%	31%	13%-58%	-0.63	100%	13.7	0%
Extra Long (11+)	11	91%	45%	21%-72%	+1.27	100%	9.0	10%

Third-down target leaders by distance



1D = first downs gained on those targets. Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

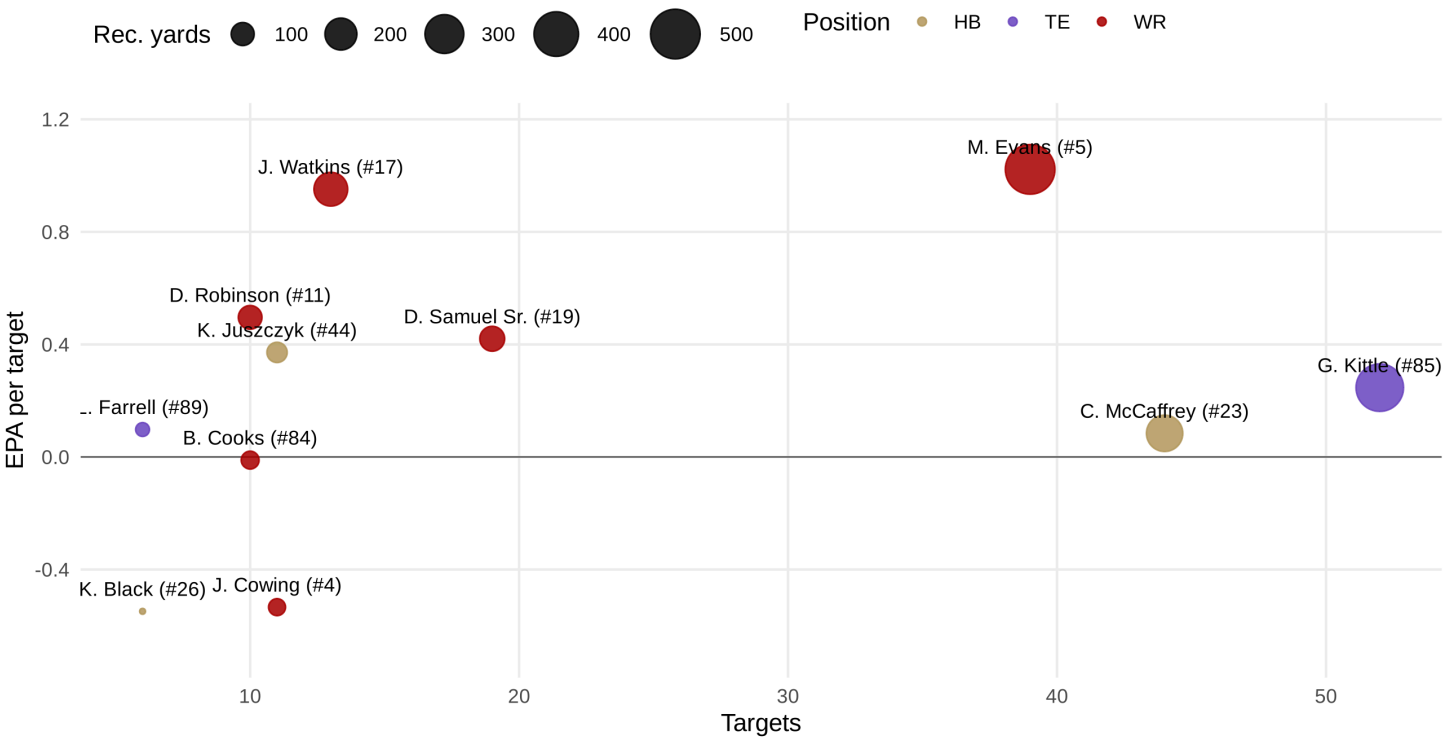
All third downs

Player	Pos	Targets	Share	Rec	Yds	1st downs	EPA/tgt	aDOT
M. Evans (#5)	WR	19	34%	12	206	12	+1.00	13.8
C. McCaffrey (#23)	HB	10	18%	5	48	4	+0.09	5.4
G. Kittle (#85)	TE	9	16%	6	67	4	+0.16	9.6
J. Cowing (#4)	WR	5	9%	2	14	1	-1.29	7.4
J. Watkins (#17)	WR	4	7%	3	85	3	+0.98	17.0
B. Cooks (#84)	WR	3	5%	0	0	0	-1.51	11.0
D. Robinson (#11)	WR	2	4%	0	0	0	-1.58	9.5
D. Samuel Sr. (#19)	WR	2	4%	2	22	2	+2.07	8.5

6. Most effective players and how they produce

Receiving volume vs. efficiency

Players with at least 5 targets

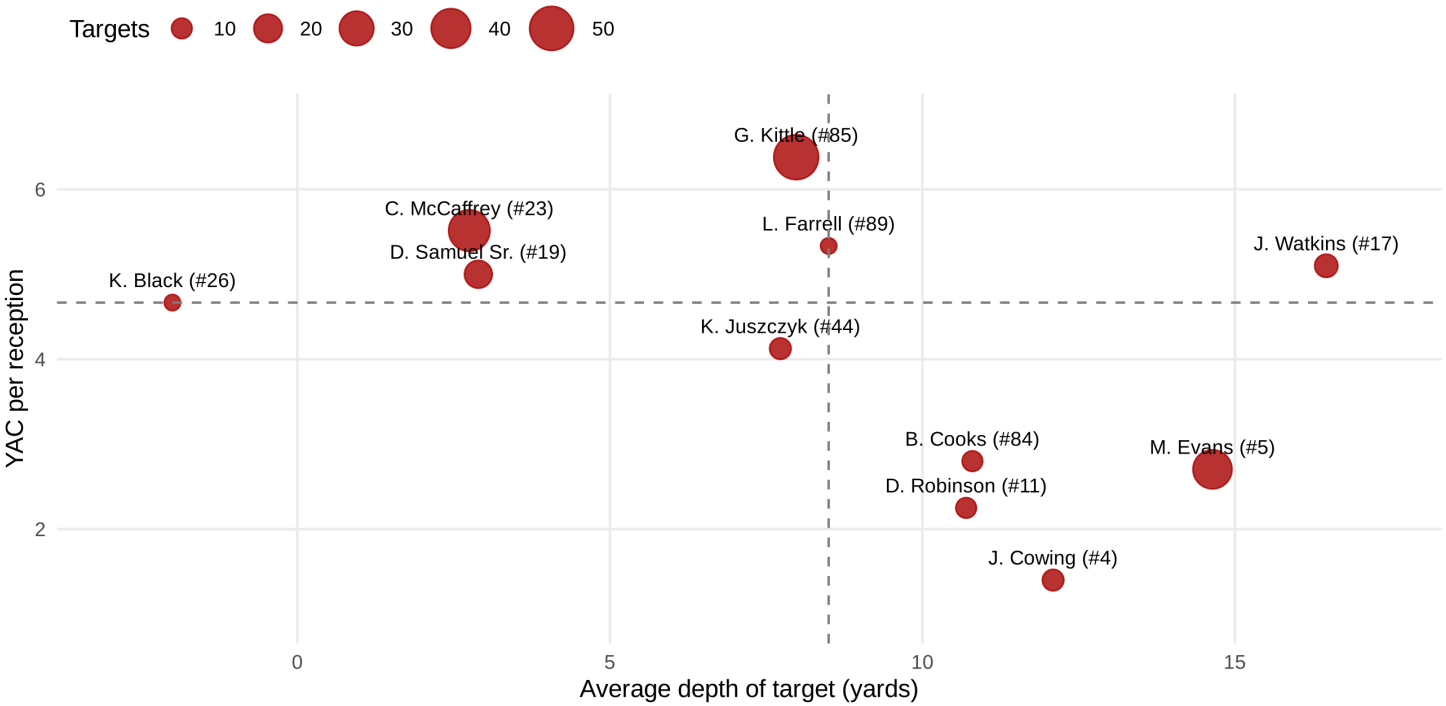


Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Player	Pos	Tgt	Share	Rec	Yds	Catch %	Y/Tgt	aDOT	YAC/Rec	EPA/Tgt	1D
G. Kittle (#85)	TE	52	23%	37	457	71%	8.8	8.0	6.4	+0.25	21
C. McCaffrey (#23)	HB	44	20%	35	262	80%	6.0	2.8	5.5	+0.08	13
M. Evans (#5)	WR	39	17%	27	500	69%	12.8	14.6	2.7	+1.02	25
D. Samuel Sr. (#19)	WR	19	8%	15	118	79%	6.2	2.9	5.0	+0.42	5
J. Watkins (#17)	WR	13	6%	10	224	77%	17.2	16.5	5.1	+0.95	9
J. Cowing (#4)	WR	11	5%	5	52	45%	4.7	12.1	1.4	-0.53	3
K. Juszczyk (#44)	HB	11	5%	8	76	73%	6.9	7.7	4.1	+0.37	6
B. Cooks (#84)	WR	10	4%	5	58	50%	5.8	10.8	2.8	-0.01	5
D. Robinson (#11)	WR	10	4%	8	106	80%	10.6	10.7	2.2	+0.50	6
K. Black (#26)	HB	6	3%	3	4	50%	0.7	-2.0	4.7	-0.55	0
L. Farrell (#89)	TE	6	3%	3	33	50%	5.5	8.5	5.3	+0.10	2
J. Brendel (#64)	OL	1	0%	1	2	100%	2.0	-1.0	3.0	-0.44	0

How receivers produce: depth of target vs. yards after catch

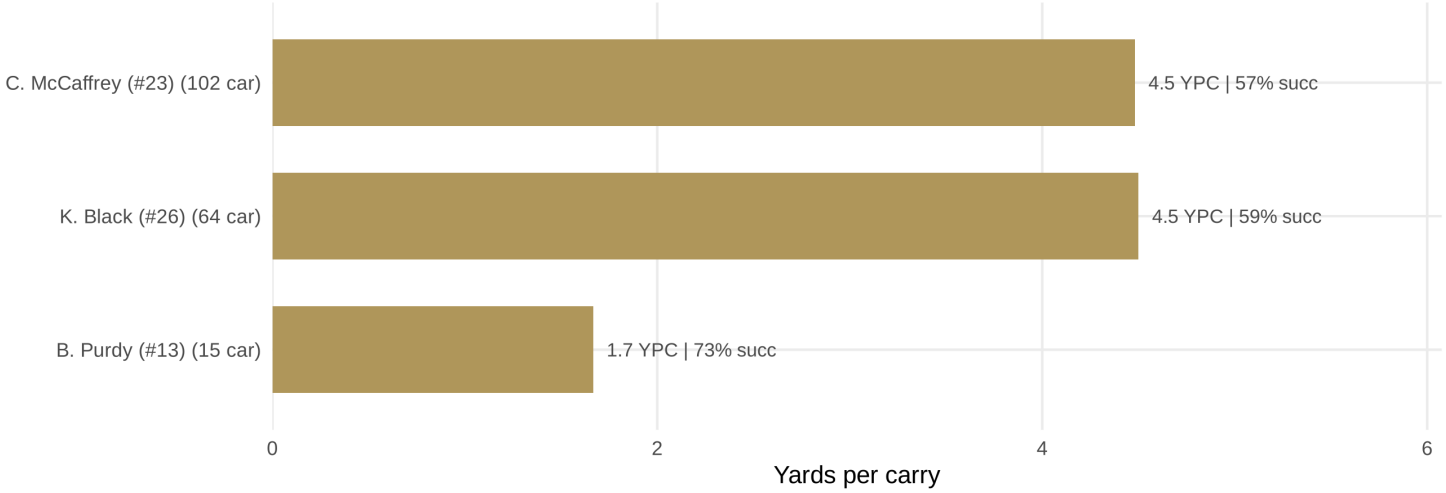
Right = targeted deeper downfield; up = more yards after the catch. Dashed = team medians



Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Rushers: yards per carry

Yards after contact is not in this data source, so carries are not split



Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Player	Carries	Share	Yds	YPC	Success	EPA/car	YBC	YACo	Explosive %	Stuff %	Top concept
C. McCaffrey (#23)	102	52%	457	4.5	57%	+0.13	n/a	n/a	13%	11%	Middle
K. Black (#26)	64	32%	288	4.5	59%	-0.06	n/a	n/a	9%	20%	Middle
B. Purdy (#13)	15	8%	25	1.7	73%	-0.02	n/a	n/a	0%	20%	Middle
D. Samuel Sr. (#19)	9	5%	78	8.7	56%	-0.30	n/a	n/a	44%	11%	Left End
J. James (#29)	4	2%	17	4.2	50%	-0.09	n/a	n/a	0%	0%	Right Guard
J. Cowing (#4)	2	1%	12	6.0	100%	+0.59	n/a	n/a	50%	0%	Left End
K. Juszczyk (#44)	1	1%	4	4.0	100%	+0.38	n/a	n/a	0%	0%	Middle

Total offensive value (rushing + receiving EPA; QB passing EPA is credited to targets):

Player	Carries	Targets	Scrimmage yds	Rush EPA	Rec EPA	Total EPA	EPA/opp
M. Evans (#5)	0	39	500	0.0	39.9	39.9	+1.02
C. McCaffrey (#23)	102	44	719	13.2	3.7	16.9	+0.12
G. Kittle (#85)	0	52	457	0.0	12.8	12.8	+0.25
J. Watkins (#17)	0	13	224	0.0	12.4	12.4	+0.95
D. Samuel Sr. (#19)	9	19	196	-2.7	8.0	5.3	+0.19
D. Robinson (#11)	0	10	106	0.0	5.0	5.0	+0.50
K. Juszczyk (#44)	1	11	80	0.4	4.1	4.5	+0.37
K. Hodge (#86)	0	1	17	0.0	1.0	1.0	+0.98

7. Passing-game context

Split	Value	Dropbacks	Share	EPA/play	Success	Y/P
Play action	No	198	77%	+0.37	59%	8.4
Play action	Yes	58	23%	+0.08	55%	6.7
Screen	No	235	92%	+0.34	59%	8.1
Screen	Yes	21	8%	-0.06	52%	6.6

Targets by depth

Depth	Targets	Share	Comp %	Y/Tgt	EPA/Tgt	INT %
Behind LOS	39	17%	85%	4.7	-0.31	3%
Short (1-9)	100	45%	79%	7.5	+0.44	1%
Intermediate (10-19)	65	29%	55%	9.7	+0.37	5%
Deep (20+)	20	9%	55%	17.9	+1.15	0%

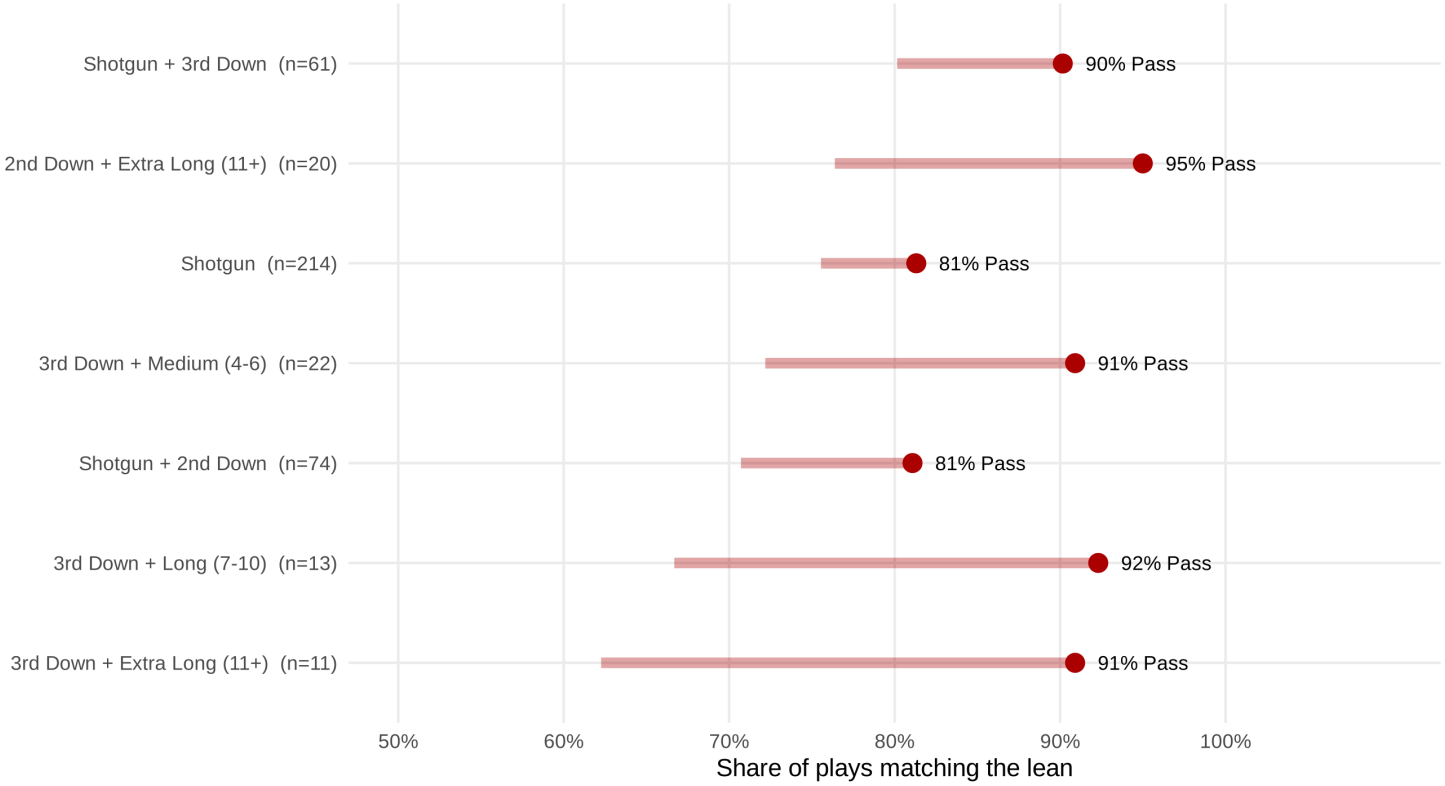
Results by defensive coverage faced (5+ dropbacks)

Coverage	Dropbacks	Share	EPA/play	Success
Unknown	256	100%	+0.30	58%

8. Pre-snap tells and predictability

Strongest pre-snap tells

Dot = observed rate; faded bar runs down to the 95% lower bound; n = plays with that cue



Data: nflverse (nflfastR pbp, NGS participation, FTN charting)

Always guessing the more common play type is right **57%** of the time. Using personnel, alignment, down and distance bucket (leave-one-out cell rates) is right **72%** of the time, a lift of **15%**.

Methodology and data notes

How the sample was built

Step	Plays
Offensive snaps in file	506
Run or pass plays (drops penalties / no-plays)	478
Downs 1-4 (drops PATs / 2-pt tries)	478
Excluding garbage time	462
Excluding QB kneels and spikes	453

WHAT NFLVERSE CANNOT COVER (OR ONLY APPROXIMATES)

- Personnel / formation / coverage / routes: nflverse participation data covers only 0% of these plays (it is typically posted after the season), so those sections are mostly empty.
- Run concepts: nflverse has no blocking-scheme charting (inside zone, duo, power ...). Section 4 groups runs by gap and direction instead (e.g. 'Left Tackle', 'Middle').
- PFF success codes (pff_OFFSUCCESS): not in nflverse. Success uses the standard 40/60/100 yardage rule only.
- EPA / EP: nflfastR's public model, not PFF's. Values are close in spirit but not identical.

- Yards after contact: not in nflverse, so the before/after-contact split for rushers is unavailable (yards per carry is shown instead).
- Receiver alignment (slot / wide / backfield): not in nflverse. 'Main align' is blank.
- Personnel is counted from the roster positions of the 11 players on the field (RB+FB, TE). A WR listed as TE on the roster counts as a TE. '(*)' = 6+ offensive linemen.
- Garbage time is approximated with win probability (second half, pre-snap WP below 0.01% or above 99.99%). Calibrated against a licensed PFF sample.
- 2-minute flag: inside the final 2:00 of either half (PFF's 4-minute flag has no equivalent).
- Pressure is NGS 'was_pressure' (yes/no); time to pressure is not available.

Column names below refer to the PFF GamePBP layout the nflverse data was mapped onto.

- **Pass play / dropback:** `pff_RUNPASS = "P"`, which includes sacks and QB scrambles. **Run:** `"R"`. Penalties and no-plays (`"X"`) are removed.
- **QB alignment:** Pistol if `pff_PISTOL` is set, otherwise Shotgun if `pff_SHOTGUN` is set, otherwise Under Center.
- **Success (standard):** gaining at least 40% of the distance on 1st down, 60% on 2nd, and 100% on 3rd/4th, or earning a first down. PFF success codes are not in nflverse, so only the standard rule is used.
- **First down:** `pff_FIRSTDOWNGAINED = 1` or yards gained ≥ distance.
- **EPA:** `pff_EXPECTEDPOINTSADDED` as provided. Passing EPA is attributed to the targeted receiver.
- **Target:** a pass play with a recorded receiver, excluding sacks and scrambles. **Receiving yards:** yards gained on completions.
- **Explosive:** runs of 10+ yards, passes of 20+ yards.
- **Field zone:** `ABSOLUTEYARDLINE` is yards to the opponent's end zone (so 1–20 is the red zone).
- **Tell:** a pre-snap cue (personnel, alignment, down, distance, or combinations) with at least 10 plays where one play type reaches 80%. The 95% lower bound uses a Wilson interval.
- **Intervals:** rates show 95% Wilson intervals; run-concept EPA shows ±1 standard error. With samples this small, most differences between groups are not statistically distinguishable; the report shows n everywhere so readers can judge.
- **Garbage time** excluded: TRUE. **2-minute drills** excluded: FALSE. **Personnel "(*)"** marks a jumbo package (an extra offensive lineman replaces a skill player); merged into the base grouping: FALSE.

Data checks:

- Player names from SF_2026_roster.csv (nflverse weekly rosters 2026)

Blank numeric cells (for example yards after catch on runs) are expected and listed for completeness.

Reproducibility: source file MD5 c20b5d30eaca12721178a3a674da44f1. Every table in this report is also saved as CSV and in one Excel workbook next to this file.